THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS TWO-CHAPTER TEST SET D

Chapters 5 & 6: **Parallel and Intersecting Lines, Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A transversal cuts two parallel lines. A corresponding angle to 104° is: 1

(a) 14° **(b)** 52° **(c)** 104° **(d)** 76°

2. A transversal cuts parallel lines. If an alternate interior angle is 41°, its pair is: 1

(a) 49° **(b)** 51° **(c)** 41° **(d)** 139°

3. The sum of co-interior angles on the same side of a transversal between parallel lines is: 1

(a) Equal **(b)** 180° **(c)** 90° **(d)** 360°

4. Angles around a point add up to: 1

(a) 180° **(b)** 360° **(c)** 90° **(d)** 720°

5. If a transversal makes alternate angles equal, the two lines are: 1

(a) equal **(b)** intersecting at 90°
(c) perpendicular **(d)** parallel

6. The sum of the first 5 odd numbers is: 1

(a) 15 **(b)** 20 **(c)** 30 **(d)** 25

7. In a magic square the centre of 1–9 is: 1

(a) 5 **(b)** 1 **(c)** 3 **(d)** 9

8. Consecutive numbers always have a sum that is: 1

(a) multiple of 4 **(b)** even **(c)** odd **(d)** multiple of 3

9. The smallest 3-digit palindrome is: 1

(a) 121 **(b)** 101 **(c)** 111 **(d)** 100

10. The number of odd numbers between 1 and 10 (inclusive) is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 10

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write one real-life example each of parallel lines and perpendicular lines. 2

12. The angles around a point are 90°, 120°, 100° and x. Find x. 2

13. A number is 9 times a digit and has digit sum 9. Name two such 2-digit numbers. 2

14. Add: 1 + 3 + 5 + 7 + 9 + 11. Which pattern is shown? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the reflex angle formed by the hands of a clock at 9 o'clock. 3

16. Explain whether 3,001 × 5 is odd or even and what its last digit is. 3

17. Three lines meet at a point forming angles x, x + 30° and 2x + 30° on a straight line. Find x. 3

OR

Explain why intersecting lines produce two pairs of equal angles. 3

18. Write 36 as a sum of consecutive odd numbers starting from 1. 3

OR

Explain why any number and its digit-reverse differ by a multiple of 9 (two digits). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) State the converse of the corresponding angles property. 5
(ii) Use it to show two lines are parallel if corresponding angles are 80° each.
(iii) Why not if one is 80° and the other 100°?

OR

(i) Three angles at a point on a straight line are x, 2x and 3x. Find x. 5
(ii) Name the largest.
(iii) Is it obtuse?

20. (i) Find the digit D if 3 × 1D5 is divisible by 9. 5
(ii) Find the number.
(iii) Check.

OR

(i) Find the sum 1 + 2 + … + 20 by pairing. 5
(ii) Explain pairing.

END OF QUESTION PAPER — Total: 40 Marks

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